

D499

Machine Learning

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A streaming service with 30 million subscribers has viewing histories but no labels, and wants to discover natural groups of viewers for marketing. The data sits in a new cloud warehouse. Which approach fits?

- A. The warehouse's built-in reporting.
- B. Supervised classification.
- C. Unsupervised clustering, such as k-means.
- D. Linear regression.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A real-estate firm with offices in five cities wants to predict each listing's sale price in dollars from size, age, and location. It has 20,000 past sales. Which task type is this?

- A. Clustering.
- B. Classification, because 20,000 records is enough for it.
- C. Classification.
- D. Regression.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which target makes a problem classification rather than regression?

- A. Grouping customers when no outcome is recorded.
- B. The number of days until a customer cancels.
- C. Whether a customer will cancel next month (yes or no).
- D. The dollar amount a customer will spend next month.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team predicting hospital readmission includes the feature “follow-up call outcome,” which is recorded after the readmission is known. The model scores 0.98 AUC. The hospital has 450 beds. What is the problem?

- A. Data leakage: the feature is not available at prediction time.
- B. The hospital is too small for machine learning.
- C. Underfitting.
- D. Nothing; 0.98 AUC always means an excellent model.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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5.

A churn model at a telecom with 6 million customers scores 99% accuracy on training data and 71% on test data. It is a very deep decision tree built by a contractor in Austin. What is happening, and what helps?

- A. Overfitting; limit tree depth, prune, or regularize.
- B. Underfitting; grow the tree deeper.
- C. Nothing; high training accuracy is all that matters.
- D. Retrain on the test data to raise its score.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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6. [TRANSFER — reversed stem / different variable]

A linear model scores 62% on training data and 61% on test data for a problem where strong models reach 90%. What is the likely issue?

- A. Overfitting.
- B. Underfitting.
- C. Nothing; matching scores always mean an ideal model.
- D. Data leakage.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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7.

A spam filter at a mid-size university flagged 120 emails as spam in one week. Of those, 90 were truly spam. There were 150 spam emails in total that week. The IT office has 40 staff. What is the filter's precision?

- A. 0.25
- B. 0.80
- C. 0.75
- D. 0.60

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A model has recall of 0.80, and there were 250 actual positives. How many actual positives did it miss?

- A. It cannot be found without precision.
- B. 50
- C. 200
- D. 80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A fraud model at a bank with 900 branches reports 99.2% accuracy. Fraud is 0.8% of transactions. The model was trained on a powerful GPU cluster. What is the most likely problem?

- A. It may predict “not fraud” every time; check recall and precision on the fraud class.
- B. It is excellent, because 99% accuracy always means a strong model.
- C. It is overfitting.
- D. It needs a bigger GPU cluster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In a data set, 95% of cases are negative. A model predicts every case negative. What are its accuracy and recall?

- A.** 5% accuracy, 100% recall.
- B.** 0% accuracy, 0% recall.
- C.** 95% accuracy, 95% recall.
- D.** 95% accuracy, 0% recall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A k-nearest neighbors model at a car insurer uses annual income (range \$20,000 to \$250,000) and number of past claims (range 0 to 5). The data came from 14 states. Before training, what should be done?

- A.** Drop the claims feature because its numbers are small.
- B.** Nothing; k-NN is never affected by feature scale.
- C.** Train a separate model for each of the 14 states.
- D.** Scale the features (standardize or min-max) so income does not dominate distance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A team with only 800 labeled records from a small lender wants a reliable estimate of model performance. Their laptop has 16 GB of RAM. Which method fits best?

- A.** Score the model on its training data.
- B.** k-fold cross-validation.
- C.** Buy more RAM first.
- D.** A single 50/50 train-test split.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A streaming service ~~with 30 million subscribers~~ has viewing histories but no labels, and wants to discover natural groups of viewers for marketing. ~~The data sits in a new cloud warehouse.~~ Which approach fits?

Target word: learning type **Decides:** No labels; goal is to find groups.

A	Camouflage	Reports summarize; they do not discover segments.
B	Confusion set	Needs labeled classes, which do not exist.
C	Correct	Clustering finds groups without labels.
D	Adjacent category	Predicts a numeric label; there is none.

2. Answer D Concept 2

A real-estate firm ~~with offices in five cities~~ wants to predict each listing's sale price in dollars from size, age, and location. It has 20,000 past sales. Which task type is this?

Target word: task type **Decides:** The target is a dollar amount.

A	Adjacent category	Clustering has no target.
B	Camouflage	Data size does not decide the task type.
C	Confusion set	Classification predicts categories.
D	Correct	Predicting a continuous number is regression.

3. Answer C Concept 2 · transfer

Which target makes a problem classification rather than regression?

Target word: target type — reverse **Decides:** A categorical target.

A	Adjacent category	Unsupervised clustering.
B	Confusion set	A numeric count usually treated as regression.
C	Correct	A two-class outcome.
D	Inverted operation	A continuous target: regression.

4. Answer A Concept 3

A team predicting hospital readmission includes the feature “follow-up call outcome,” which is recorded after the readmission is known. The model scores 0.98 AUC. ~~The hospital has 450 beds.~~ What is the problem?

Target word: problem **Decides:** The feature is recorded after the outcome.

A	Correct	Future information inflates the score.
B	Camouflage	Bed count does not explain the score.
C	Inverted operation	The score is suspiciously high, not low.
D	Absolute language	High scores can come from leakage.

5. Answer A Concept 4

A churn model at a telecom ~~with 6 million customers~~ scores 99% accuracy on training data and 71% on test data. It is a very deep decision tree ~~built by a contractor in Austin~~. What is happening, and what helps?

Target word: diagnosis and fix **Decides:** Large gap between training and test.

A	Correct	The tree memorized training data.
B	Inverted operation	More depth worsens overfitting.
C	Absolute language	Test accuracy measures real performance.
D	Confusion set	That leaks the test set and hides the problem.

6. Answer B Concept 4 • transfer

A linear model scores 62% on training data and 61% on test data for a problem where strong models reach 90%. What is the likely issue?

Target word: diagnosis — reverse case **Decides:** Both scores low and close.

A	Inverted operation	Overfitting shows a big train-test gap.
B	Correct	The model is too simple to capture the pattern.
C	Absolute language	They can match at a low level.
D	Confusion set	Leakage inflates scores.

7. Answer C Concept 5

A spam filter at a mid-size university flagged 120 emails as spam in one week. Of those, 90 were truly spam. There were 150 spam emails in total that week. ~~The IT office has 40 staff~~. What is the filter's precision?

Target word: precision **Decides:** TP = 90 of 120 flagged.

A	Inverted operation	30 / 120 is the share of flags that were wrong.
B	Wrong stem number	120 / 150 mixes flagged and actual counts.
C	Correct	$90 / 120 = 0.75$.
D	Confusion set	90 / 150 is recall.

8. Answer B Concept 5 • transfer

A model has recall of 0.80, and there were 250 actual positives. How many actual positives did it miss?

Target word: false negatives from recall **Decides:** $TP = 0.80 \times 250 = 200$.

A	Absolute language	Recall and the positive count are enough.
B	Correct	$250 - 200 = 50$ missed.
C	Inverted operation	That is the number caught.
D	Wrong stem number	Reads the recall as a count.

9. Answer A Concept 6

A fraud model at a bank with 900 branches reports 99.2% accuracy. Fraud is 0.8% of transactions. The model was trained on a powerful GPU cluster. What is the most likely problem?

Target word: problem **Decides:** 99.2% equals the non-fraud share.

A	Correct	Accuracy equals the majority rate, so it may catch nothing.
B	Absolute language	Imbalance makes accuracy misleading.
C	Confusion set	No train-test gap is given.
D	Camouflage	Hardware does not fix a metric problem.

10. Answer D Concept 6 · transfer

In a data set, 95% of cases are negative. A model predicts every case negative. What are its accuracy and recall?

Target word: compute both metrics **Decides:** All negatives right; no positives caught.

A	Inverted operation	That would be predicting every case positive.
B	Dropped element	Drops the correct negatives from accuracy.
C	Confusion set	Recall counts only actual positives.
D	Correct	Right on the 95% negatives; catches no positives.

11. Answer D Concept 7

A k-nearest neighbors model at a car insurer uses annual income (range \$20,000 to \$250,000) and number of past claims (range 0 to 5). The data came from 14 states. Before training, what should be done?

Target word: preprocessing step **Decides:** Distance-based model with very different ranges.

A	Dropped element	Removes a useful feature instead of scaling it.
B	Absolute language	Distance depends directly on scale.
C	Camouflage	Does not solve the scale problem.
D	Correct	Puts features on comparable scales.

12. Answer B Concept 8

A team with only 800 labeled records from a small lender wants a reliable estimate of model performance. Their laptop has 16 GB of RAM. Which method fits best?

Target word: evaluation method **Decides:** Small labeled data set.

A	Dropped element	Drops held-out testing, which overstates performance.
B	Correct	Every record is tested once, steadying the estimate.
C	Camouflage	Memory does not change the evaluation design.
D	Adjacent category	One split on small data gives a noisy estimate.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Supervised vs. unsupervised learning	Supervised learning trains on labeled outcomes; unsupervised learning finds structure (such as clusters) in unlabeled data.	Reaching for classification when there are no labels.	"No labels," "discover groups."
2	Classification vs. regression	Predicting a category is classification; predicting a continuous number is regression.	Choosing by data size or model name rather than by the target's type.	The target: yes/no or class vs. dollars, days, degrees.
3	Data leakage	Features must be available at prediction time; information from the future or from the label leaks and inflates scores.	Celebrating a near-perfect score without checking feature timing.	A feature recorded after the outcome, plus a suspiciously high score.
4	Overfitting vs. underfitting	High training score with much lower test score = overfitting (reduce complexity or regularize); low scores on both = underfitting (add capacity or features).	Adding complexity to an overfit model.	Training vs. test scores side by side.
5	Precision and recall	Precision = $TP / (TP + FP)$, the share of flagged items that are right; recall = $TP / (TP + FN)$, the share of real positives caught.	Swapping the denominators.	"Of those flagged ..." vs. "of all actual ..."
6	Accuracy with imbalanced classes	When one class is rare, accuracy can be high while the model catches none of the rare class; use recall, precision, F1, or PR curves.	Trusting 99% accuracy on a 1% event.	Accuracy close to the majority-class rate.
7	Feature scaling	Distance-based methods (k-NN, k-means, SVM) need features on similar scales, or large-range features dominate.	Feeding raw dollars and small counts into a distance model.	Features with very different ranges and a distance-based algorithm.
8	Cross-validation	k-fold cross-validation rotates the test fold so every record is tested once, giving a steadier performance estimate on small data.	Evaluating on training data or on one small split.	Few labeled records and a need for a reliable estimate.

Course scope (public description): Machine Learning presents the end-to-end process of investigating data through a machine learning lens. Topics covered include: supervised and unsupervised learning algorithms, features that best represent data, commonly-used machine learning algorithms, and methods for evaluating the performance of machine learning algorithms.